Module 7

Comprehending Cloud Networking

Module Objectives

- Understand various cloud services and network components.
- Review cloud network troubleshooting techniques and tools.

Lesson 7.1

Cloud Networking Concepts

Cloud+ CV0-004 Certification Exam Objective:

- 1.3 Explain Cloud Networking Concepts.

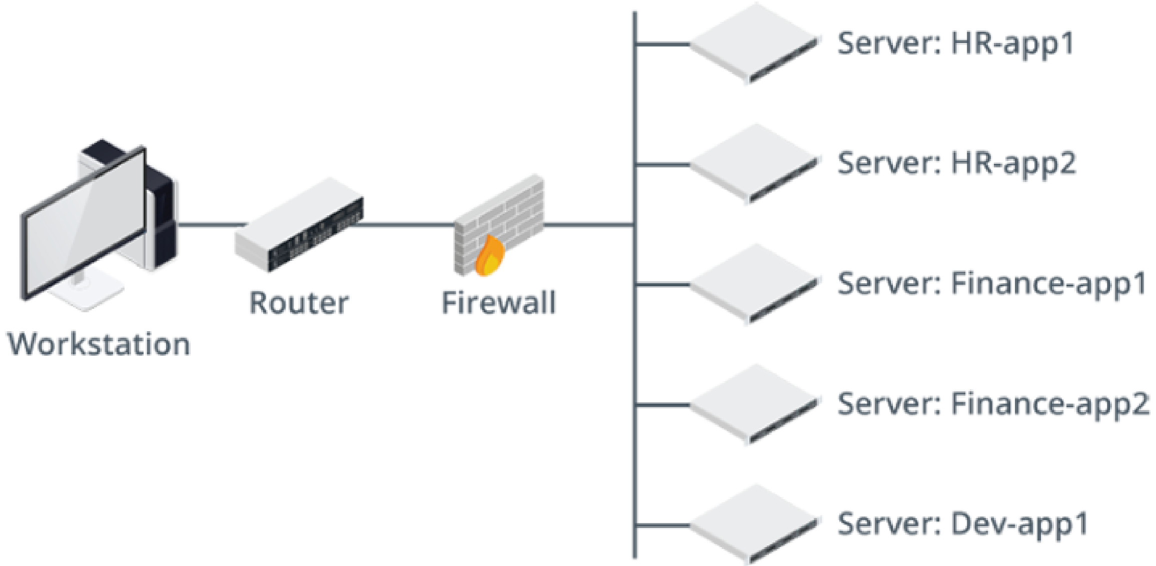
Cloud Networking Concepts

- Networking Concepts
- Network Flow Diagram
- Public and Private Cloud Connections
- Virtual Private Network Concepts
- Virtual Private Network Designs
- Secure Shell
- Secure Shell Key-Based Authentication
- IPsec Security
- Dedicated Connections

Networking Concepts

- Secure cloud connectivity options
- Internet Gateway
- NAT Gateway

Network Flow Diagram



This network flow diagram displays the relationships between compute environments. Note that all environments are in a single segment. (Images © 123RF.com)

Public and Private Cloud Connections

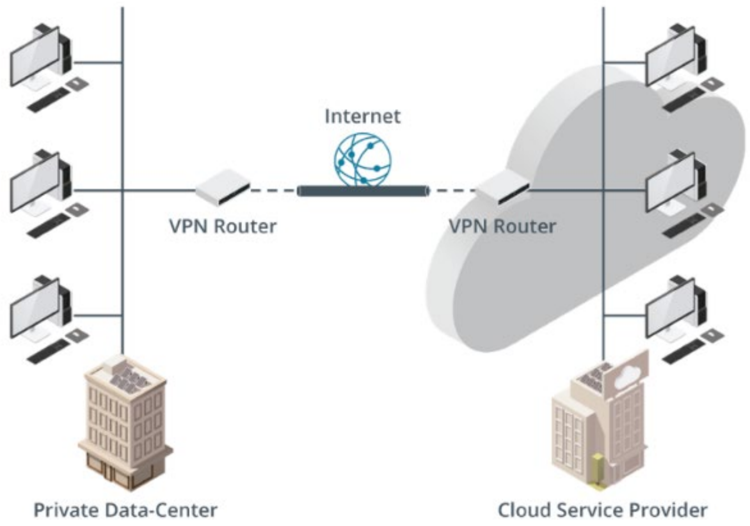
- Necessity of VPN encryption
- Data protection via VPN
- VPN protocols and security

Virtual Private Network Concepts

- Protects network connections between on-premises resources and cloud services
- Threats
- Mitigation

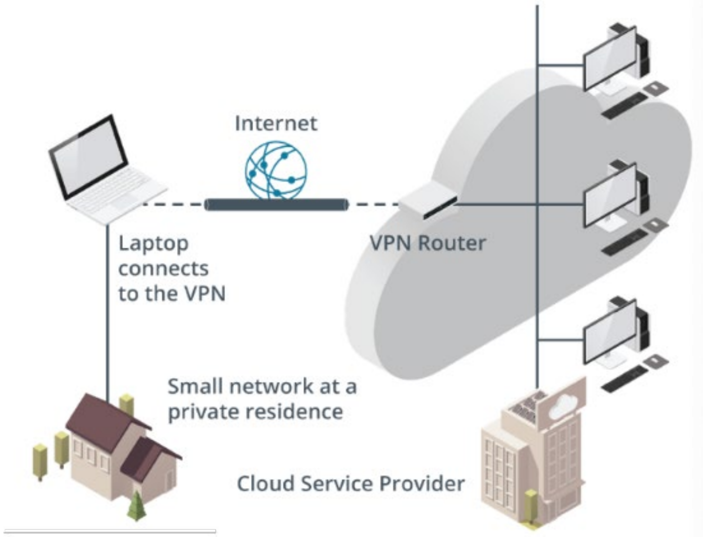
Virtual Private Network Designs

Site-to-Site VPN



A VPN connection from the data center to a CSP. (Images © 123RF.com)

Point-to-Site VPN



Home-to-cloud service provider point-to-site VPN. (Images © 123RF.com)

Secure Shell

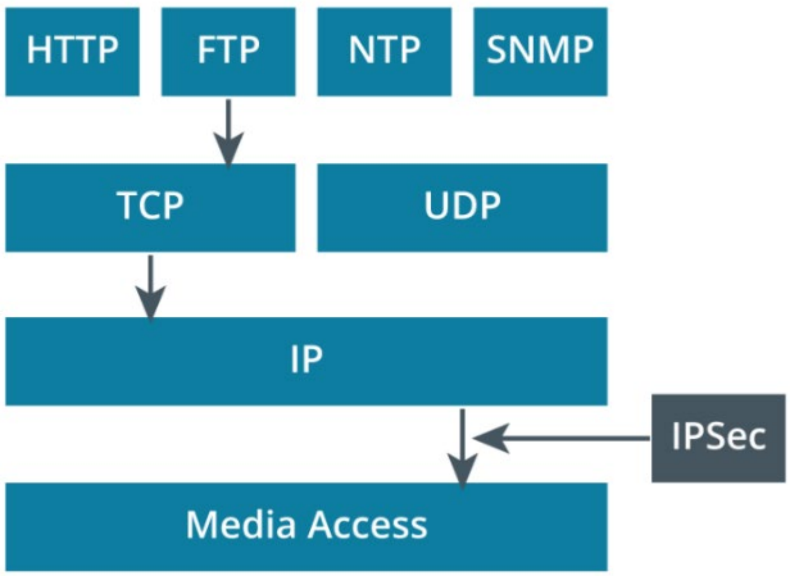
- Administrator logs on to local Linux workstation.
- Connection passes through company's network to ISP.
- Connection traverses the Internet.
- Connection enters CSP's network and data center.
- Connection reaches VPC and target cloud VM.

Secure Shell Key-Based Authentication

- Generate key pair:
  - *$ ssh-keygen*
- Send public key to remote VM:
  - *$ ssh-copy-id {remote Linux cloud VM}*
- Test connection:
  - *$ ssh {remote Linux cloud VM}*
- No password prompt; authentication via key exchange.

IPsec Security

- Network layer
  - Transmission Control Protocol/Internet Protocol (TCP/IP) stack
- Modes
  - Transport mode
  - Tunnel mode



The TCP/IP stack with IPsec and the FTP protocol encrypted.

Dedicated Connections

- Concerns with Public Internet
- Dedicated Connections
  - GCP: Dedicated Interconnect
  - AWS: Direct Connect
  - Azure: ExpressRoute

Lesson 7.2

Network Functions

Cloud+ CV0-004 Certification Exam Objective:

- 1.3 Explain cloud networking concepts.

Network Functions

- Load Balancers
- Cloud Firewalls
- Web Application Firewalls

Load Balancers

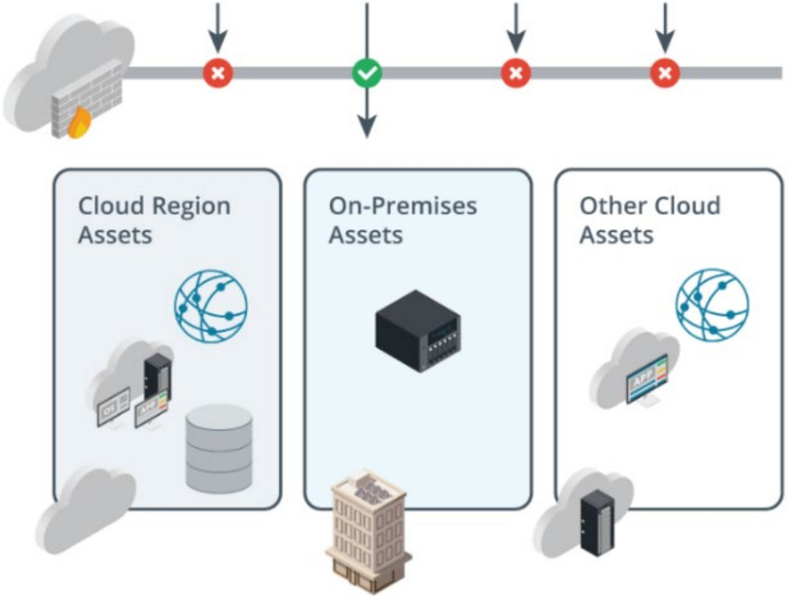
- Recognizing application load balancers
- Recognizing network load balancers
- Recognizing gateway load balancers

Cloud Firewalls

- Stateful
  - Transmission Control Protocol (TCP)
- Stateless

Web Application Firewalls (WAFs)

- Operate at Layer 7 (Application layer)
- Apply virtual patches to protect applications
- Criteria for rules



A web application firewall protects a company's cloud and on-premises assets. (Images © 123RF.com)

Lesson 7.3

Network Components

Cloud+ CV0-004 Certification Exam Objective:

- 1.3 Explain cloud networking concepts.

Network Components

- Switches
- Network Segmentation and Subnetting
- Microsegmentation
- Routing Tables
- Border Gateway Protocol
- Virtual Network Interface Cards

Switches

- Operate at Layer 2 of the OSI model
- Physical switches
- Virtual switches

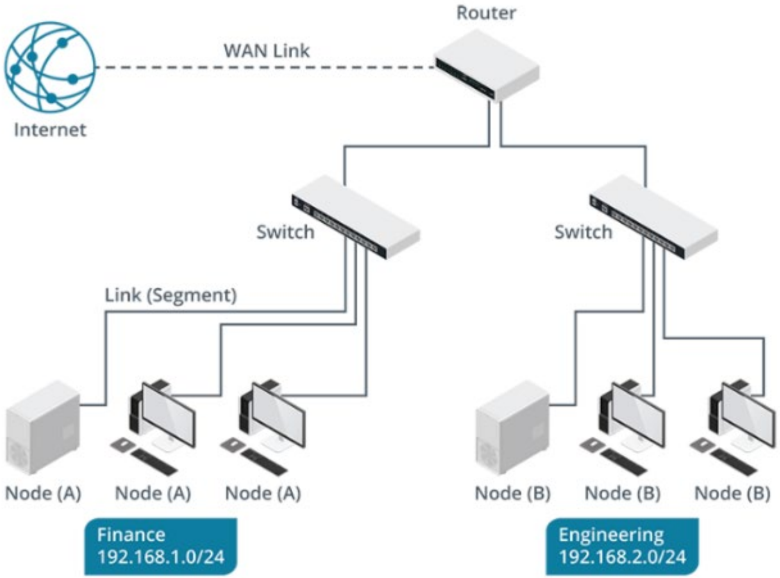
- Features
  - VLAN Tagging
  - Port Mirroring
  - Quality of Service (QoS)

Network Segmentation and Subnetting

- Single network with subnets
- Multiple virtual networks configured as peers
- Multiple virtual networks in a hub-and-spoke topology

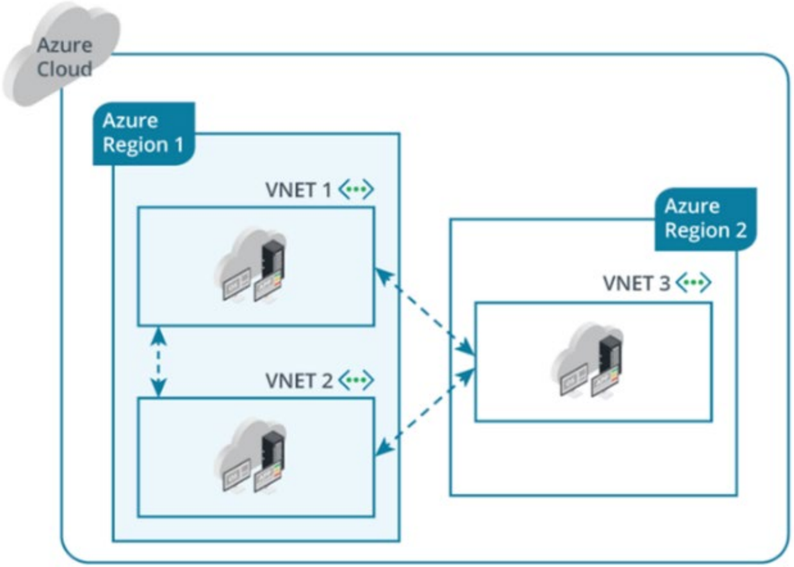
Network Segmentation and Subnetting

Single Network with Subnets



Network Segmentation and Subnetting

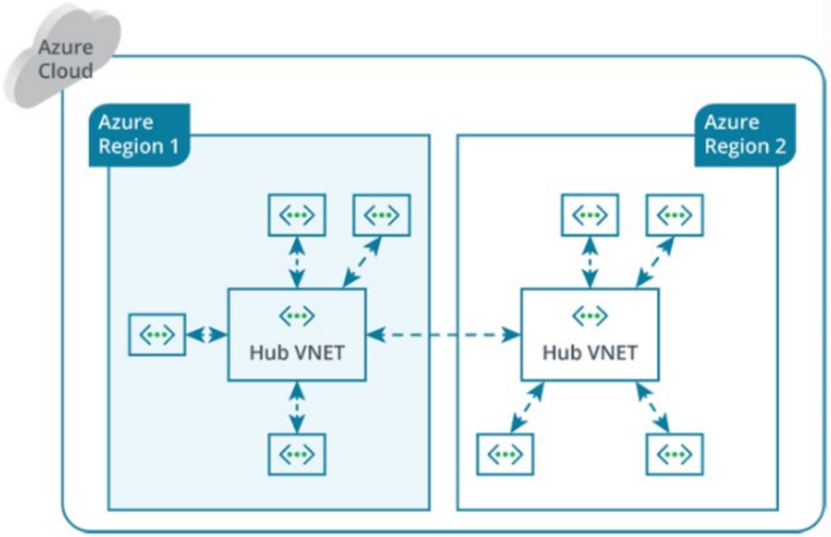
Multiple Virtual Networks Configured as Peers



Three virtual networks configured as peers of each other. (Images © 123RF.com)

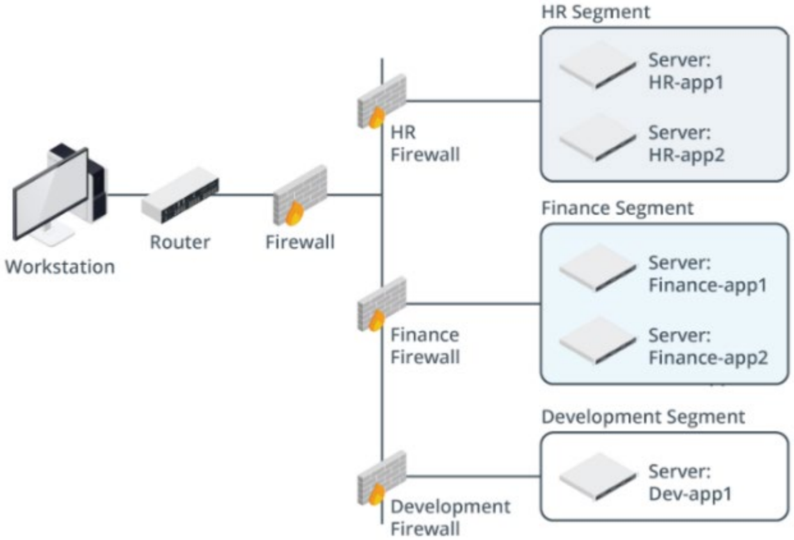
Network Segmentation and Subnetting

Multiple Virtual Networks in a Hub-and-Spoke Topology



Multiple virtual networks in a hub-and-spoke design. Note that the two hub virtual networks are peers. (Images © 123RF.com)

Microsegmentation



This network flow diagram displays how microsegmentation isolates various compute environments, such as HR, Finance, and Development applications. (Images © 123RF.com)

- Visibility
- Granularity
- Dynamic

Routing Tables

- Interpreting route tables
- Static routing tables
- Dynamic routing tables

Border Gateway Protocol (BGP)

- Common routing protocols
  - Open Shortest Path First (OSPF)
  - Enhanced Interior Gateway Routing Protocol (EIGRP)
  - Border Gateway Protocol (BGP)
- Autonomous Systems (AS)

Virtual Network Interface Cards (vNICs)

The table lists the maximum number of network interfaces per instance type and the maximum number of private IPv4 and IPv6 addresses per network interface.

Instance Type	Maximum Network Interfaces	Private IPv4 Addresses per Interface	IPv6 Addresses per Interface
c3.2xlarge	4	15	15
c3.4xlarge	8	30	30
c3.8xlarge	8	30	30

Lesson 7.4

Network Services

Cloud+ CV0-004 Certification Exam Objective:

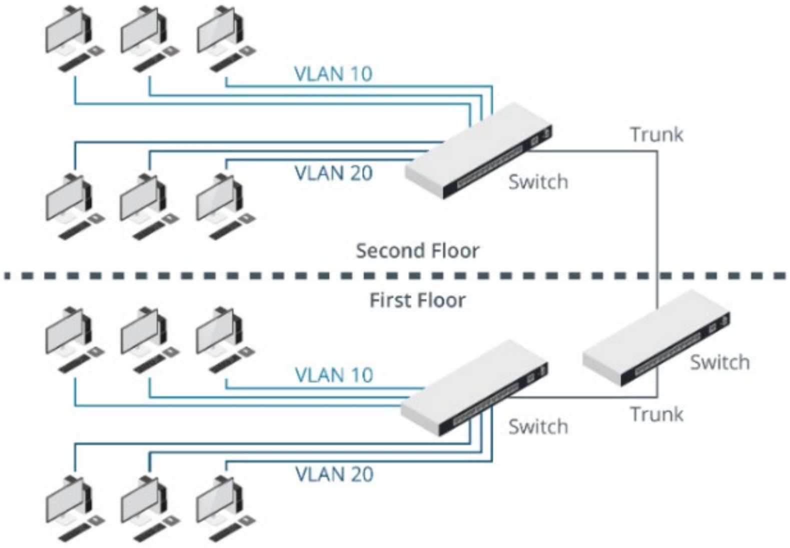
- 1.3 Explain cloud networking concepts.

Network Services

- Virtual LAN Technologies
- Virtual Extensible LAN Settings
- Software-Defined Networking
- Virtual Private Clouds
- Virtual Private Cloud Designs
- Network Address Translation in Virtual Private Clouds
- Transit Gateway in the Cloud
- Content Delivery Networks

Virtual LAN Technologies

- Add Layer 2 switch ports to specific VLANs using software rules
- Isolate traffic within switches using VLAN-specific tags
- Support up to 4094 networks

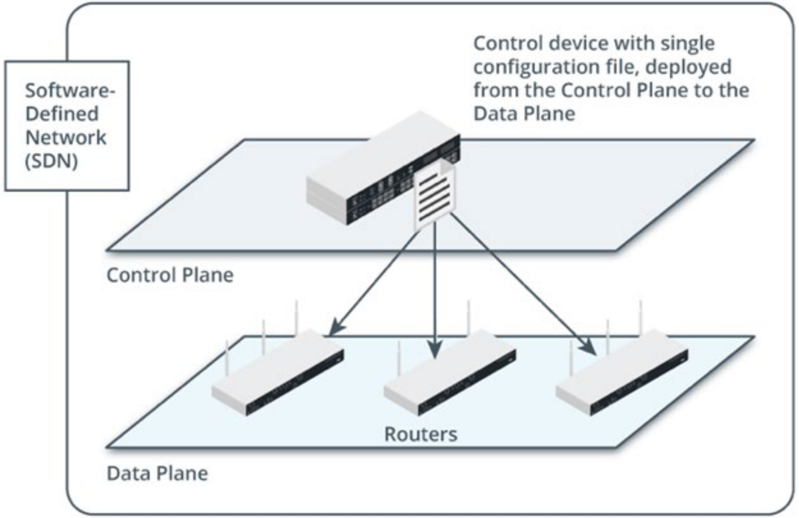


Some nodes are in VLAN-10 and some are in VLAN-20. (Images © 123RF.com.)

Virtual Extensible LAN Settings

- Network stretching
- Traffic mirroring
- Geneve

Software-Defined Networking



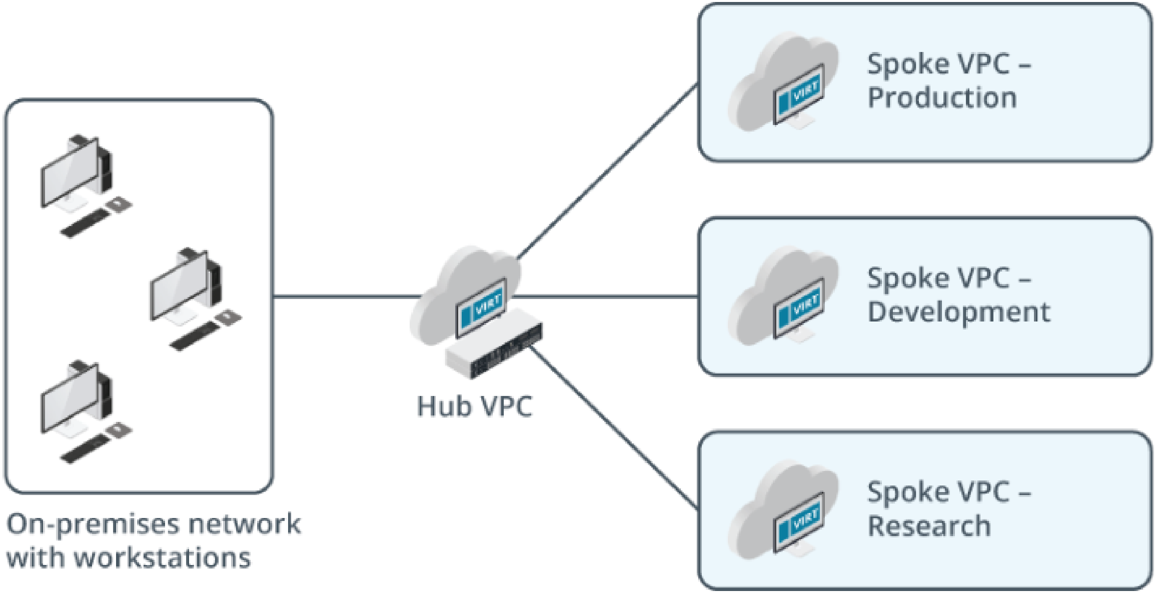
Data plane devices are managed by a control plane device. (Images © 123RF.com)

- Data plan
- Control plane

Virtual Private Clouds (VPCs)

- Achieve greater security within a public cloud
- Implement a single-tenant structure within a CSP's multi-tenant environment
- Components
  - Subnets
  - VLANs
  - VPNs
  - Instances
  - Web apps

Virtual Private Cloud Designs



Hub-and-spoke VPC design, with separate Production, Development, and Research spoke VPCs connected to a central "hub" VPC. (Images © 123RF.com)

Network Address Translation in Virtual Private Clouds

- Client network address translation
- Cloud network address translation

CompTIA

Transit Gateway in the Cloud

- Acts as a hub to manage and direct traffic between network areas
- Connections
  - Virtual Private Clouds (VPCs)
  - On-premises networks via VPNs
  - On-premises networks via dedicated connections
  - Software-Defined WAN (SD-WAN) connections

Content Delivery Network (CDN)

- How content delivery networks work
- Handling high traffic and failures

Content Delivery Network with data centers around the world. (Images © 123RF.com.)

Lesson 7.5

Cloud Network Services

Cloud+ CV0-004 Certification Exam Objective:

- 1.3 Explain cloud networking concepts.

Cloud Network Services

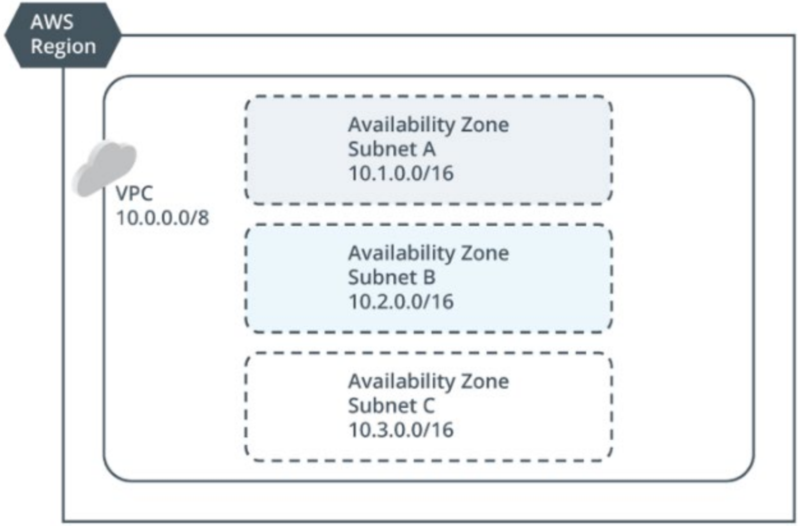
- Name Resolution
- Dynamic Host Configuration Protocol
- IP Address Tracking
- Network Time Protocol

Name Resolution

- Converts human-friendly names to IP addresses for network services
- Critical for both on-premises and cloud-based resources
- Cloud DNS services
- Securing name resolution

Dynamic Host Configuration Protocol

- Configuration information
- Configuration methods
- DHCP lease generation process
- IP addressing in VPCs



Cloud region with VPC further subdivided into subnets. (Images ©123RF.com)

IP Address Tracking

- Manage IP address allocations
  - Create network map
  - Track IP addresses

Network Time Protocol (NTP)

- Time synchronization
- Importance
- Cloud environments
- Configuration
- Network Time Security (NTS)

Lesson 7.6

Troubleshoot Network Issues

Cloud+ CV0-004 Certification Exam Objective:

- 6.2 Given a scenario, troubleshoot network issues.

Troubleshoot Network Issues

- Standard Network Troubleshooting Tools
- DNS Service Availability
- IP Address Service Availability
- DHCP Service Availability
- Network Time Protocol Service Availability
- Hypertext Transfer Protocol Issues
- Network Configuration Issues
- Network Configuration Latency Issues
- Network Bandwidth and Throughput Issues
- Device Misconfiguration Issues
- Protocol Incompatibility and Deprecation
- Routing Issues
- Network Address Translation
- Switching Issues
- Packet Capture and Scanning Utilities

Standard Network Troubleshooting Tools

- Ping
  - Success
  - Fail: REQUEST TIMED OUT
  - Fail: DESTINATION HOST UNREACHABLE
- Traceroute / tracert
  - Confirm the trace passes through the correct routers

DNS Service Availability

- DNS resource records
- Name resolution

IP Address Service Availability

- Incorrect IP configurations
- IP addresses
- Route information

DHCP Service Availability

- DHCP scope exhaustion
- Network overlap

Network Time Protocol Service Availability

- Purpose
- Configuration
- Troubleshooting time synchronization issues

Hypertext Transfer Protocol Issues

- Troubleshooting Failed HTTP Connections
- HTTP Status Codes:
  - 1xx, 2xx, 3xx, 4xx, 5xx
  - 403 Forbidden: Authorization problem.
  - 404 Not Found: Resource can't be found.
  - 500 Internal Server Error: Nonspecific web service issue.
  - 502 Bad Gateway

Network Configuration Issues

- Potential error sources
- Common causes
- Troubleshooting steps

Network Configuration Latency Issues

- Latency-sensitive application
- Troubleshooting latency
- Quality of Service (QoS) issues

Network Bandwidth and Throughput Issues

- Bandwidth and throughput issues
- Setting maximum transmission unit sizes
- Network connectivity and performance issues

Device Misconfiguration Issues

- Proxy configurations
- Network connection information

Protocol Incompatibility and Deprecation

- Virtual LAN (VLAN) protocol incompatibilities
- Hypertext transfer protocol (HTTP) vs. HTTPS requirements
- Protocol deprecations
- Non-encrypted protocols

Routing Issues

- Missing routes
- Route misconfiguration issues
- Virtual private cloud (VPC) issues

NAT Troubleshooting

Class	Reserved IP Address Range
Class A	10.0.0.0/8
Class B	172.16.0.0/12
Class C	192.168.0.0/16

- No internet access
  - Device offline
  - NAT tables inaccurate
- Internal IP addresses are mishandled by NAT over a VPN
  - Verify NAT protocols
  - Verify NAT configuration
  - Implement a specific NAT for VPN solution

Switching Issues

- VLAN configurations
- Misconfigured tags
- Access and trunk ports

Packet Capture and Scanning Utilities

- Packet captures
- Network scanners

Module Summary

- Cloud networks virtualize familiar physical devices (routers, firewalls, switches).
- Enable network traffic control, performance management, and integration with on-premises components.
- Standard network concepts and troubleshooting methods remain useful.
- Secure connections using VPNs and other technologies.
- Use firewalls, routers, and segmentation to isolate services.
- Software-defined networking (SDN) simplifies management, scaling, and quick deployments.